

LEARNING SOFT SKILLS WITH AN AI-BASED SIMULATION ROLE-PLAY: A LITERATURE REVIEW

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Abstract

In the past five years, concurrent progress in algorithms, big data, and hardware combined with efforts of numerous actors have resulted in a sudden outburst of fruitful competition in Artificial Intelligence (AI) developments. This technology has been utilized to give rise to a new generation of Virtual Learning Environments (VLE) that rely on Intelligent Tutoring Systems (ITS) powered by AI. As a result, the more than thirty-year-old field of Artificial Intelligence in Education (AIED) has become widely popularized.

However, the development of generative AI tools such as ChatGPT and DALL·E has sparked debates within the educational sphere, particularly regarding legal and ethical considerations. In response, schools in New York and universities like Sciences Po in Paris have opted to ban the use of ChatGPT, prioritizing principles of academic integrity and knowledge authenticity. Nevertheless, AI technologies, including bots, AI tutors, and simulators, have garnered significant interest for their ease of access and potential ability to streamline learning processes, reduce costs, and facilitate personalized instruction.

The concept of learning with generative AI has emerged as a focal point of current research, investigating how AIED can provide teachers with predictive tools while providing students with support mechanisms such as recommendation systems and chatbots, and feedback channels, including virtual agents.

In an era marked by rapid digital transformation and the increasing automation of jobs, the acquisition of soft skills has never been more crucial in cultivating an adaptable workforce and fostering career development. This article delves into the intersection of artificial intelligence and soft skills development, offering a literature review on how AI-driven methodologies can enhance learning and the acquisition of these essential competencies. It explores the potential of AI to bridge the gap between theoretical knowledge and practical application, emphasizing the restoration of a human touch in training and the enhancement of soft skills. The article focuses on immersive simulation role-plays powered by AI, providing a controlled and secure environment for trainees to improve their communication, leadership, and teamwork abilities.

Moreover, as a prelude to our ongoing Erasmus+ project, ROLEPL-AI, this article delves into an examination of contemporary literature findings on the impact of AI from a cognitive psychology perspective, considering factors such as metacognitive processes, socio-emotional interactions, and ergonomic guidelines for interactions. These insights will underpin the development of ROLEPL-AI, our experimental AI-based simulation role-play learning environment, specifically tailored to Vocational Education and Training (VET) in the field of tourism.

Keywords: Generative AI, Virtual Learning Environment (VLE), Intelligent Tutoring System (ITS), Vocational Education and Training (VET), tourism, Large Language Models (LLM), ChatGPT, AI, GPT.

1 INTRODUCTION

With the popularisation of generative Artificial Intelligence (AI), new techniques and tools are being explored in computer-assisted learning and Virtual Learning Environments (VLE). Intelligent Tutoring Systems (ITS) can leverage the readily available abilities of a new generation of chatbots that use Large Language Models (LLM) to engage more naturally with the learners. This is especially promising in the field of soft skills training. Soft skills, encompassing attributes like emotional intelligence, adaptability, teamwork, and communication, are integral aspects of personal and professional success [1], [2]. Indeed, simulating credible human interactions in an open scenario may become a matter of integrating a specifically trained and fine-tuned LLM module.

In this context, the Erasmus+ project ROLEPL-AI was initiated to explore and assess the efficacy of AI-based simulations for soft skills training within the tourism sector, to help bridge the gap in the existing literature regarding this subject. Three expert knowhows are put together: LLM training and use, development of VLEs and immersive environments, and tourism Vocational Education and Training (VET). Learners should eventually be able to interact with specially trained LLMs that take the role of professionals and customers in a simulated roleplay situation in a virtual campus. In addition, the same trained LLMs should give feedback and advice on the learning material and strategy as virtual tutors. This would give this new generation of AI systems the roles of both roleplay agents and tutors.

According to the European Commission's 2018 definition, "AI refers to systems that display intelligent behaviour by analysing their environment and taking action – with some degree of autonomy – to achieve specific goals." [3] This includes technologies like machine learning, natural language processing, computer vision, and expert systems, with applications in every sector, from robotics to consumer chatbots to health, and of course to education. AI has become a component of the educational landscape, starting with simple in-school computer-assisted learning tools and extending with home edutainment or personal learning apps.

The popularisation of generative AI among the general public has reached the education sector as well. This happened not just among the teachers but also among the learners themselves. In a study, around 50% of respondents reported regular use of generative AI tools like ChatGPT in their academic work [4]. This raises multiple questions: will education globally benefit from this new tool by allowing for efficient and secure learning? Will identified and unexpected drawbacks be addressed in time, be them about data privacy, critical thinking skills, dependency to AI-based tools, or algorithmic bias, among others? Therefore, there is a need for rigorous research and analysis to understand the complex relationship between AI and education and its implications for teaching methods, curriculum design, and student learning experiences.

With this goal in mind, this article aims to investigate the relationship between AI and education, with a focus on the emergence of Artificial Intelligence in Education (AIED) as a crucial concept in current educational discussions. By drawing on insights from cognitive psychology, educational theory, and technological innovation, we aim to clarify the potential benefits and challenges associated with the integration of AI in educational settings. Specifically, we will examine the impact of AI on learning processes and especially the development of soft skills – learning *with* AI as opposed to learning *about* AI [5]. In fact, the cultivation of AI literacy among students and educators, would also be an interesting lead to explore, as gaining a deeper understanding of a tool often facilitates mastery over it. AI literacy involves acquiring knowledge and understanding about the principles, applications, and consequences of AI [6]. This includes learning about the various components and applications of AI, as well as understanding its societal impacts.

By conducting a thorough examination of existing literature and empirical data, this article will study the influence of AI-driven methodologies on enhancing teaching and learning methodologies. Our main goal is to enrich the ongoing discourse surrounding AI in education by offering valuable insights and perspectives. These findings aim to guide future research, shape policy frameworks, and inform pedagogical approaches in the digital era. Through a critical analysis of the opportunities and obstacles posed by AI technologies, we intend to foster a nuanced understanding of their capacity to transform the landscape of education.

2 METHODOLOGY

The research criteria consisted in prioritising experimental articles, which put theories to the test, while partially excluding theoretical articles, when based on a thought experiment only. The total number of resources reviewed was 130.

Academic research websites were explored using keywords such as learning, soft skills, artificial intelligence, and then further browsing was performed according to the citations in the articles. The search was carried out by segmentation and subsequently by opening, which allowed to find articles based on solid theories on the main subjects: learning and soft skills.

As far as the time periods are concerned, a restrictive selection was made for research into AI, learning and soft skills, excluding articles before 2020. For the definition of learning, the articles used are from 2000 onwards.

3 RESULTS

3.1 Results from cognitive science

Significant findings emerged regarding the impact of AI interaction on cognitive processes among students. Lin [7] explained that environments integrating AI interaction positively influenced cognitive faculties such as problem-solving, logical reasoning, and collaboration. Moreover, AI has been recognized for its potential in bolstering metacognitive processes among learners, including activities like self-assessment and inquiry [8], [9].

Fu et al. [10] underscored the role of AI equipped with automatic responses and feedback in supporting learners who might refrain from seeking clarification from instructors in traditional settings. Similarly, Kim et al. [11] advocated for the utilization of AI as a collaborative learning agent, fostering peer-like interactions and feedback within classrooms. These studies accentuate the active engagement of learners with AI, positioning them as dynamic participants rather than passive recipients.

The literature emphasizes that active involvement in the learning process correlates with heightened engagement, improved learning outcomes, and better retention compared to passive information reception [12], [13], [14]. However, there exists a notable gap in the literature concerning comparative studies between passive and active learning with AI.

Furthermore, Nazaretsky et al. [15] developed an AI algorithm aimed at supporting teachers in enhancing cognitive processes. This algorithm provides tailored learning analyses, enabling personalized instruction aligned with students' competency levels. Their research underscores the efficacy of collaboratively developed AI algorithms in modern science classrooms within blended learning environments.

Studies predominantly position AI as a diagnostic tool for enhancing feedback provision, vital for learning performance and goal pursuit [16]. Motivation levels significantly influence learning strategies, with intrinsic motivation positively correlated with effective strategy employment [17]. Chatbot-based learning has emerged as a promising avenue, fostering intrinsic motivation among students [18], [19], [20].

Moreover, Huang et al. [21] investigated an AI system designed to stimulate intrinsic motivation, revealing its positive impact on learning performance, engagement, and motivation levels. However, discussions on AI and cognitive interaction necessitate addressing potential cognitive overload. Luo et al. [22] highlighted the varying benefits of AI coaches in training scenarios, suggesting a combination of AI and human coaches as the most effective approach to mitigate information overload.

Efforts to reduce cognitive overload in e-learning environments align with multimedia theory guidelines [23], grounded in cognitive load theory [24]. While no direct research assesses the efficacy of these principles in AI-based environments, studies in e-learning and virtual reality contexts have demonstrated their effectiveness in reducing cognitive load [25], [26]. Prioritizing the avoidance of cognitive overload when integrating AI in learning settings entails employing AI as a supportive tool and ensuring adherence to multimedia theory guidelines, thus optimizing learning experiences.

3.2 Specificities of soft skills training

The recognition of soft skills' importance extends beyond specific industries, permeating work environments and educational spheres alike. While traditional AI research has predominantly focused on technical skills, contemporary studies are increasingly emphasizing interpersonal competencies.

This shift aligns with the growing interest in leveraging AIED to nurture non-epistemic competencies [27], including metacognition, self-regulated learning [28], emotional intelligence [29], social interaction skills [30], and motivation [31].

The current discourse surrounding soft skills training underscores the pivotal role of emotions in learning performance, particularly the positive affective states [29], [32]. Advanced Learning Technologies (ALTs) are increasingly utilized to induce positive emotions, with a focus on developing emotion-aware systems [33]. Hwang [34] demonstrated how AI interventions effectively mitigate anxiety related to learning subjects by analysing emotions and cognitive performance to deliver personalised feedback. Similarly, McLaren [35] observed enhanced learning outcomes through interactions with polite web-based tutors, highlighting the significance of emotionally engaging learners, which fosters affective arousal and emotional bonds within learning environments [36].

Incorporating AI-virtual agents into learning environments facilitates interactions that simulate real-world scenarios, offering valuable learning experiences. Dai et al. [37] showcased the potential of AI-virtual agents in simulating challenging situations, garnering positive feedback from instructors. Additionally, interacting with social robots instead of humans has an adverse influence on learning performance [38], even when participants did not report a negative attitude towards technology. Recognizing these nuances, efforts are directed toward inducing positive emotions in ALTs through proactive and reactive approaches, aimed at constructing positive environments and monitoring user states [29].

Research on aware systems has led to the development of tools like FATiMA, designed to imbue virtual characters with social and emotional intelligence [39]. While primarily applied outside educational contexts, these advancements hold potential for enhancing learning experiences. Furthermore, AI interactions exert both indirect and direct influences on learning outcomes, particularly in the development of soft skills, which are increasingly valued in today's job market due to concerns about technological job displacement [40].

In educational settings, strategies for cultivating soft skills are evolving, with role-playing emerging as a prevalent method. The effectiveness of AI-based simulation training in enhancing soft skills is well-documented, particularly in industries like healthcare and hospitality [41], [42]. For instance, Liaw et al. [43] demonstrated significant improvements in communication skills following AI-based simulation training in healthcare. Simulation-based training provides a safe environment for learners to practice and refine their skills, with studies indicating positive impacts on skill mastery [44], [45].

While empirical research in specific sectors like hospitality remains limited [46], projects such as the Erasmus+ Hotel Academy have shown promise in fostering essential competencies through simulation-based learning. The project aimed to enhance collaboration, communication, and strategic thinking skills through desktop and VR-based scenarios, yielding positive feedback from participants [47]. As the demand for soft skills continues to grow, leveraging AI-driven simulation training holds immense potential for preparing individuals for the challenges of the modern workforce.

3.3 Learning with AI-based simulations

When exploring the dynamics of human-new technology interactions, a human-centered approach is necessary for the development of these technologies. Current research in AI interaction design relies on traditional user interface design guidelines, with Nielson's heuristics [48], [49] serving as foundational principles for creating user-centered interfaces. However, as AI interfaces evolve, they may pose challenges to existing guidelines due to their dynamic nature and personalized responses, as noted by Amershi et al. [50].

As the discrepancy between traditional interface guidelines and AI interfaces becomes evident, recent studies have sought to address this issue. Fu et al. [10] demonstrated the impact of AI speech recognition accuracy and social presence on user interaction with AI, highlighting the importance of accurate and responsive AI interfaces. Drawing from affordance theory [51], which underscores the importance of accurate speech recognition in enhancing social presence, Fu et al.'s findings suggest implications for developing AI environments for learning.

Moreover, Jung et al. [52] identified a correlation between engagement, presence, and learning persistence, emphasizing the significance of perceived teaching support presence in distant learning modalities. This underscores the importance of investigating the perceived presence of teaching support in AI applications to facilitate continued user engagement.

To address the evolving landscape of AI interfaces, Amershi et al. [50] proposed a set of guidelines for designing Human-AI interactions (Table 1), aiming to enhance usability and usefulness. However, it is acknowledged that AI interfaces are diverse, adapted to specific needs and contexts, as emphasized by Carvalho et al. [53]. Consequently, while general principles are valuable, they must be complemented by context-specific guidelines, such as Mayer's guidelines elaborated for learning environments [23].

Table 1. Human-AI interaction design guidelines by Amershi & al. (2019).

	AI Design Guidelines		Example Applications of Guidelines
Initially	G1	Make clear what the system can do. Help the user understand what the AI system is capable of doing.	[Activity Trackers, Product #1] "Displays all the metrics that it tracks and explains how. Metrics include movement metrics such as steps, distance traveled, length of time exercised, and all-day calorie burn, for a day."
	G2	Make clear how well the system can do what it can do. Help the user understand how often the AI system may make mistakes.	[Music Recommenders, Product #1] "A little bit of hedging language: 'we think you'll like'."
During interaction	G3	Time services based on context. Time when to act or interrupt based on the user's current task and environment.	[Navigation, Product #1] "In my experience using the app, it seems to provide timely route guidance. Because the map updates regularly with your actual location, the guidance is timely."
	G4	Show contextually relevant information. Display information relevant to the user's current task and environment.	[Web Search, Product #2] "Searching a movie title returns show times in near my location for today's date"
	G5	Match relevant social norms. Ensure the experience is delivered in a way that users would expect, given their social and cultural context.	[Voice Assistants, Product #1] "[The assistant] uses a semi-formal voice to talk to you - spells out 'okay' and asks further questions."
	G6	Mitigate social biases. Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.	[Autocomplete, Product #2] "The autocomplete feature clearly suggests both genders [him, her] without any bias while suggesting the text to complete."
When wrong	G7	Support efficient invocation. Make it easy to invoke or request the AI system's services when needed.	[Voice Assistants, Product #1] "I can say [wake command] to initiate."
	G8	Support efficient dismissal. Make it easy to dismiss or ignore undesired AI system services.	[E-commerce, Product #2] "Feature is unobtrusive, below the fold, and easy to scroll past...Easy to ignore."
	G9	Support efficient correction. Make it easy to edit, refine, or recover when the AI system is wrong.	[Voice Assistants, Product #2] "Once my request for a reminder was processed I saw the ability to edit my reminder in the UI that was displayed. Small text underneath stated 'Tap to Edit' with a chevron indicating something would happen if I selected this text."
	G10	Scope services when in doubt. Engage in disambiguation or gracefully degrade the AI system's services when uncertain about a user's goals.	[Autocomplete, Product #1] "It usually provides 3-4 suggestions instead of directly auto completing it for you"
	G11	Make clear why the system did what it did. Enable the user to access an explanation of why the AI system behaved as it did.	[Navigation, Product #2] "The route chosen by the app was made based on the Fastest Route, which is shown in the subtext."
Over time	G12	Remember recent interactions. Maintain short term memory and allow the user to make efficient references to that memory.	[Web Search, Product #1] "[The search engine] remembers the context of certain queries, with certain phrasing, so that it can continue the thread of the search (e.g., 'who is he married to' after a search that surfaces Benjamin Bratt)"
	G13	Learn from user behavior. Personalize the user's experience by learning from their actions over time.	[Music Recommenders, Product #2] "I think this is applied because every action to add a song to the list triggers new recommendations."
	G14	Update and adapt cautiously. Limit disruptive changes when updating and adapting the AI system's behaviors.	[Music Recommenders, Product #2] "Once we select a song they update the immediate song list below but keeps the above one constant."
	G15	Encourage granular feedback. Enable the user to provide feedback indicating their preferences during regular interaction with the AI system.	[Email, Product #1] "The user can directly mark something as important, when the AI hadn't marked it as that previously."
	G16	Convey the consequences of user actions. Immediately update or convey how user actions will impact future behaviors of the AI system.	[Social Networks, Product #2] "[The product] communicates that hiding an Ad will adjust the relevance of future ads."
	G17	Provide global controls. Allow the user to globally customize what the AI system monitors and how it behaves.	[Photo Organizers, Product #1] "[The product] allows users to turn on your location history so the AI can group photos by where you have been."
	G18	Notify users about changes. Inform the user when the AI system adds or updates its capabilities.	[Navigation, Product #2] "[The product] does provide small in-app teaching callouts for important new features. New features that require my explicit attention are pop-ups."

4 CONCLUSIONS

This literature review listed the results of studies conducted on relevant fields connected to learning soft skills with an AI-based simulation role-play, from computer-assisted learning to cognitive science, including specificities of soft skills teaching. This is part of the theoretical preparation for the Erasmus+ ROLEPL-AI project that will see the development and assessment of an immersive environment comprising an LLM trained for role-play and tutoring of tourism soft skills training.

Results in cognitive science, primarily the theories regarding motivation, cognitive load, and self-regulation, are the foundational principles with which any educational system should be designed. They ensure that the learner has the highest likelihood of meaningfully engaging with the content.

The content itself chosen for this review is soft skills training, which happens to be the sector that will be the least affected by this most recent crisis in job automation linked to generative AI. Research shows that simulation-based learning is adapted for this content, while highlighting that more studies are needed.

Finally, the chosen medium of learning, a virtual simulation that uses AI-powered role-play, is a good fit for its learning purpose and content. The digital platform coupled with parametrizable generative AI allows for a great range of variations and thus possible adaptations to adhere to the principles drawn from research in cognitive science and in soft skills training. Generative AI itself provides support for learning by fostering learner autonomy with feedback and tutoring, while strengthening cognitive processes of the learners. Using emotions is also a dimension to explore carefully, as it facilitates learning if employed along recommended guidelines.

Given the limited availability of studies on AI-based learning utilizing generative AI, the outcomes discussed are primarily theoretical predictions, validated in related fields, or observed in a few practical instances. Additional research is necessary to conclusively affirm these effects, a gap that the Erasmus+ ROLEPL-AI project aims to address. In pursuit of this objective, we will draw upon the best practices and design guidelines outlined in this article.

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